

FEW-SHOT ACOUSTIC SYNTHESIS WITH MULTIMODAL FLOW MATCHING (FLAC)

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ABSTRACT

The paper introduces FLAC (Few-shot flow-matching acoustic synthesis), a probabilistic framework unlike traditional deterministic methods for generating acoustically consistent Room Impulse Responses (RIRs) in unseen environments using sparse information. Existing approaches require dense recordings and retraining for novel scenes, while the prior few-shot methods [1–3] remain deterministic. Hence, they fail to model the uncertainty in sparse acoustic observations due to material or geometric interdependencies. FLAC addresses these limitations by employing a conditional generative model based on flow matching and diffusion transformers to synthesize distributions of plausible RIRs in a multimodal context.

The proposed architecture includes three key components: a variational autoencoder (VAE) for compact latent representation of RIR waveforms, a multimodal conditioning module for encoding geometric, spatial, and acoustic cues, and a diffusion transformer that utilizes rectified flow matching with field velocity to model latent acoustic distributions.

Additionally, to evaluate scene consistency, the author introduces AGREE (Acoustic-Geometry Embedding), a CLIP-style dual-encoder framework that aligns geometry and acoustics within a shared latent embedding space as vectors. The goal is to build an embedding space where matching acoustics and geometry vectors are close together while unrelated vectors are far apart.

Experimental results demonstrate that FLAC delivers excellent performance in both seen and unseen environments. Remarkably, the model outperforms existing eight-shot methods using a single reference RIR, while maintaining strong geometry-consistent synthesis and realistic acoustic diversity. [4–6]

1. INTRODUCTION

When recreating acoustic aspects, we expect them to be scene consistent for an enjoyable and immersive experience, especially in VR and augmented environments. The acoustic properties of a room are encapsulated by Room Impulse Responses (RIRs), which capture the complex interactions between geometry, materials, and source-listener positions. Previous neural acoustic fields require extensive data and retraining, prompting the need for innovative approaches with minimal data and no retraining.

The sparse observation setting of FLAC leads to the inherent ambiguity in RIR, which is addressed using probabilistic multimodal methods as opposed to a single deterministic outcome, generating plausible RIRs. FLAC is claimed to be the first application of flow matching for explicit RIR synthesis and also introduces AGREE, a joint acoustic-geometry embedding framework. It demonstrates the ability to generalize across new source-receiver locations in known rooms, as well as in entirely unseen environments, surpassing xRIR and other baselines.

2. METHODS

2.1. VAE

The purpose of the Variational Autoencoder (VAE) is to transform high-dimensional Room Impulse Response (RIR) waveforms into a compact latent representation, $\mathbf{z}_0$, which preserves the acoustic information while reducing computational load. The VAE contains an encoder that processes RIR waveforms through four convolutional blocks that use strided convolutions, channel expansion and downsampling using ResNet blocks and Snake activation functions [7].

The decoder maps the latent representation back to the waveform and mirrors the encoder using convolutions, which are weight-normalized, with the output bounded by tanh to match the RIR amplitude range. After the VAE compresses the target RIR waveform into a latent representation $\mathbf{z}_0$, FLAC uses scene information like acoustic, spatial and geometric context through a multimodal conditioning module to generate realistic latent RIR representations.

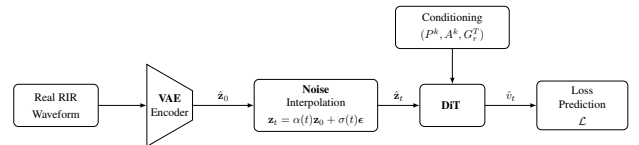


Fig. 1. Training pipeline.

2.2. Latent Flow Matching

After the target RIR is encoded into a latent vector $\mathbf{z}_0$, FLAC is trained using the rectified flow matching formulation [8, 9], where $\mathbf{z}_0$ is linearly interpolated with Gaussian noise, where

the timestep $t \in [0, 1]$ controls the noise level. Timesteps are sampled by drawing $\alpha \sim \mathcal{N}(-1.2, 2^2)$ and mapping it to t using a sigmoid. The noisy latent at timestep t is:

$$\mathbf{z}_t = (1 - t)\mathbf{z}_0 + t\epsilon$$

The model does not directly predict the clean RIR; instead, $u(\mathbf{z}_t, t, \tau)$ is trained to predict the velocity field, which tells the model in which direction the noisy latent should move to become a realistic RIR.

$$\mathbf{v}_t = \frac{d\mathbf{z}_t}{dt} = \epsilon - \mathbf{z}_0$$

2.3. Diffusion Transformer

After the VAE encodes the RIR into latent space, the diffusion transformer receives $\mathbf{z}_t$, together with acoustic, geometric, and spatial conditioning information, and predicts a velocity field that guides the noisy latent toward a realistic RIR, as shown in Fig. 1. A 1D convolution is followed by a linear layer that maps between the VAE latent space and the transformer embedding dimension. Each transformer block follows a fixed sequence:

- Self-attention with Rotary Positional Embedding (RoPE) [10]
- Cross-attention over conditioning tokens
- Feedforward Network (FNN) layers and residual networks

These components incorporate acoustic, spatial, and geometric context, with residual connections applied inside each sub-layer through cross-attention. Finally, the model consists of 12 transformer blocks with 8 attention heads and a hidden width of 256. We train with a learning rate of 5×10^{-5} , AdamW optimizer [11], and batch size of 64.

2.4. Inference

It employs classifier-free guidance [12] where the guided velocity prediction is given by

$$\hat{u}(\mathbf{z}_t, t, \tau) = u(\mathbf{z}_t, t, \emptyset) + \omega [u(\mathbf{z}_t, t, \tau) - u(\mathbf{z}_t, t, \emptyset)]$$

where $\omega > 0$ controls the conditioning strength, and $u(\mathbf{z}_t, t, \emptyset)$ denotes the unconditional prediction.

As shown in Fig. 2, during inference, FLAC starts from pure Gaussian noise ϵ and repeatedly applies the guided velocity field. The predicted velocity field is then integrated backward from $t = 1$ to $t = 0$, until a realistic latent representation of RIR is obtained. The final latent representation $\mathbf{z}_0$ is passed through the VAE decoder to reconstruct the synthesized Room Impulse Response (RIR) waveform.

2.5. AGREE: Acoustic–Geometry Embedding

AGREE is a contrastive framework inspired by CLIP [13], which connects text and images, whereas AGREE connects Room Impulse Response (RIR) and 3D room geometry in order to evaluate geometry consistency.

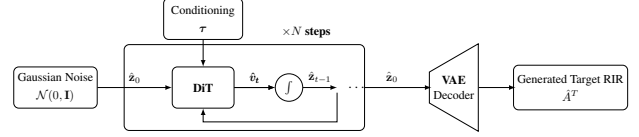


Fig. 2. Inference pipeline

It builds an embedding space where matching acoustics and geometry are near, and unrelated acoustics and geometry are far apart. The audio encoder is fine-tuned from the pre-trained VAE encoder used in FLAC and the geometric encoder, which is DINOv3 ViT-S/16 fine-tuned on panoramic depth maps. The depth values are projected into 3D coordinates, then expressed in the receiver frame, i.e., the receiver is taken as the origin, followed by source position subtraction.

Each encoder is followed by a linear projection that maps encoder outputs into a shared embedding dimension. AGREE enables zero-shot cross-modal retrieval and geometry-aware evaluation metrics, complementing traditional acoustic metrics.

3. EVALUATION

AcousticRooms (AR) Dataset [2] is used to train the VAE, multimodal conditioning modules, and diffusion transformer, while evaluation is done on HearingAnything-Anywhere (HAA) [14], which are real-world recordings used to test simulation-to-real transfer. Perceptual quality is evaluated using the relative error T_{60} , the clarity error based on C_{50} and the Early Decay Time (EDT) error. Scene-consistency metrics are evaluated by computing audio-to-audio recall, Frechet distance FD_G . With 8 shots, FLAC matches or beats Diff-RIR on perceptual metrics and xRIR in most cases. Moreover, a listening study with 46 participants on 14 unseen AR scenes in which FLAC was preferred by participants. This provides evidence that the proposed framework FLAC performs better compared to the existing techniques.

4. CONCLUSIONS

FLAC is a flow-matched multimodal few-shot acoustic synthesis method that generates Room Impulse Responses (RIRs). Unlike previous deterministic approaches, FLAC delivers a probabilistic output that addresses the inherent ambiguity of the room. Experimental results on datasets show state-of-the-art performance in novel environments, even with a single reference RIR. The paper also introduces AGREE, a joint embedding space connecting room geometry and acoustics for zero-shot retrieval and geometry-consistency evaluation. Despite its strong performance, FLAC still relies heavily on simulated training environments and currently focuses on monaural Room Impulse Responses.

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All technical content, implementations, analysis, and conclusions presented in this work were carried out and verified by the author, with the help of prior contributions by other scholars.

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